

Polar linear canonical ridgelet transform

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ABSTRACT

In recent years, the linear canonical transform has been widely applied in signal and image processing, demonstrating its powerful value as a mathematical tool. However, with the rapid development of information technology, the increasing degrees of freedom in signals have led to a growing demand for directional representations. Traditional methods are insufficient in effectively capturing the directional features of high dimensional signals. To address these challenges, the polar linear canonical ridgelet transform (PLCRT) is proposed in this paper. Firstly, the theoretical foundation of the PLCRT is thoroughly explored, with a systematic exposition of its definition, fundamental properties, inversion formula. Subsequently, the convolution theorem, correlation theorem, uncertainty principle for the PLCRT are derived. Finally, the potential application of the PLCRT in image edge detection is presented.

1. Introduction

With the continuous development of signal processing technology, effectively extracting directional and local features of high dimensional signals in the spatiotemporal domain has become a key challenge in many application fields. Traditional Fourier and wavelet transforms demonstrate outstanding performance in analyzing signal frequency and local features, paving the way for their application in signal and image processing, geophysics, astrophysics, quantum mechanics, and medicine [1–7]. However, these transforms exhibit limitations when handling non-stationary high dimensional signals, especially since the scaling property of wavelet transforms is inherently isotropic, making it difficult to effectively capture complex geometric features such as edges and corners in high-dimensional signals [8–14].

To address these challenges, various mathematical tools have been proposed, such as the ridgelet transform [8,9], polar wavelet transform (PWT) [10,11], polar Stockwell transform (PST) [12,13], linear canonical transform (LCT) [15–20], linear canonical wavelet transform (LCWT) [21], linear canonical ridgelet transform [22]. These tools have been widely applied in image compression, texture analysis, and edge detection [23,24]. However, with the increasing demand for describing directional features of signals in image processing, these methods have not yet achieved optimal results in the directional analysis of high-dimensional signals [20,21,25]. They are limited in their ability to accurately describe the directional features of images, as well as per-

form rotation, projection, and orientation analysis for high-dimensional signals across multiple angles and directions [16,26]. This limitation makes it challenging to precisely analyze and process high-dimensional signals in predetermined directions, necessitating the development of new mathematical methods to better address these challenges, which has become a research focus [17,27–29].

To overcome the aforementioned limitations, this paper proposes a new transform method called the polar linear canonical ridgelet transform (PLCRT), combining the advantages of the Radon transform [30], PWT [10,11] and the LCT. The PLCRT establishes a unified mathematical framework for the multi-scale, multi-angle, and multi-directional analysis of high-dimensional signals. The core idea of the PLCRT is to integrate the characteristics of the Radon transform, PWT and the LCT. Firstly, the Radon transform is used to project the signal onto various directions to extract directional features. Then, in the Radon domain, the PWT is applied to adjust the scale and frequency characteristics of the signal, enhancing its analysis capabilities at different resolutions and extracting ridge like features. Finally, the LCT is used to further process information in the ridgelet domain. The PLCRT with a high degree of freedom and flexibility, enabling precise analysis of the direction, scale, and frequency of high-dimensional signals, as well as better handling signals with rotational symmetry or local directional features in the polar coordinate system. As a new mathematical tool, the PLCRT demonstrates broad application potential and provides strong support for solving complex information problems [10,11,16,17,26–29].

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The remainder of this paper is organized as follows: Section 2 presents the necessary preliminaries. In Section 3, the theoretical foundation of the PLCRT is developed, including its definition, fundamental properties, inversion formula, uncertainty principle, and associated convolution and correlation theorems. Section 4 discusses potential applications of the PLCRT. Section 5 concludes the paper.

2. Preliminaries

In this section, some background knowledge related to this paper will be provided, including Radon transform [30], polar wavelet transform (PWT) [8,9], two-dimensional linear canonical transform (2D-LCT) [15].

2.1. Radon transform

The Radon transform dates back to 1917 with the seminal work of Johann Radon to construct a bivariate function from its integrals over all straight lines in a plane [30].

Definition 1. For any bivariate function $f(\mathbf{t}) \in L^2(\mathbb{R}^2)$, the Radon transform is defined as

$$R[f](\mathbf{x}, \gamma, \beta) = \int_{\mathbb{R}^2} f(\mathbf{t}) \delta(\Lambda_{\gamma, \beta} \cdot \mathbf{t} - \mathbf{x}) d\mathbf{t}, \quad (1)$$

where $\mathbf{x} = (x_1, x_2)^T \in \mathbb{R}^2$, $\mathbf{t} = (t_1, t_2)^T \in \mathbb{R}^2$, and δ denotes the well known Dirac function, and $\Lambda_{\gamma, \beta} = \begin{pmatrix} \cos \gamma & \sin \gamma \\ \cos \beta & \sin \beta \end{pmatrix}$ is the orthogonal matrix in the direction γ and β , $\sin(\beta - \gamma) = 1$.

2.2. Polar wavelet transform

The literature [10,11] systematically propose the theoretical framework of the PWT, which is constructed based on non-single mode Lie groups. Building upon this foundation, the literature [10,11] further investigate the associated square integrable representations and uncertainty principles, from which they derive the admissibility conditions required for the validity of the transform.

Definition 2. For any $f(\mathbf{t}) \in L^2(\mathbb{R}^2)$, the PWT is defined as

$$W_f(\mathbf{b}, a, \alpha) = \int_{\mathbb{R}^2} f(\mathbf{t}) \overline{\psi_{\mathbf{b}, a, \alpha}(\mathbf{t})} d\mathbf{t}, \quad (2)$$

where the wavelet kernel $\psi_{\mathbf{b}, a, \alpha}$ satisfies

$$\psi_{\mathbf{b}, a, \alpha}(\mathbf{t}) = \frac{1}{a} \psi \left(\frac{R_\alpha(\mathbf{t} - \mathbf{b})}{a} \right),$$

and R_α stands for rotation parameter given by

$$R_\alpha(\mathbf{t}) = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} t_1 \\ t_2 \end{pmatrix}, \quad \forall \mathbf{t} \in \mathbb{R}^2,$$

where $a \in \mathbb{R}^+$, $\mathbf{b} \in \mathbb{R}^2$, $\alpha \in [0, 2\pi)$.

2.3. Two dimensional linear canonical transform (2D-LCT)

The LCT is a class of linear integral transforms with three degrees of freedom and is a generalized form of the Fourier transform and the fractional Fourier transform. In recent years, many new theories have been proposed by researchers [15–20,26,28].

Definition 3. Given a matrix $M = (A, B; C, D)$ and $AD - BC = 1$, the 2D-LCT of any $f(\mathbf{t}) \in L^2(\mathbb{R}^2)$ is defined as

$$L^M[f(\mathbf{t})](\xi) = \begin{cases} \int_{\mathbb{R}^2} f(\mathbf{t}) \mathcal{K}^M(\mathbf{t}, \xi) d\mathbf{t}, & B \neq 0 \\ \sqrt{D} e^{i \frac{CD\xi^2}{2}} \delta(\mathbf{t} - D\xi), & B = 0 \end{cases}, \quad (3)$$

and the kernel $\mathcal{K}^M(\mathbf{t}, \xi)$ is given by

$$\mathcal{K}^M(\mathbf{t}, \xi) = \frac{1}{2\pi B} e^{j \left[\frac{A}{2B} \mathbf{t}^T \mathbf{t} - \frac{1}{B} \mathbf{t}^T \cdot \xi + \frac{D}{2B} \xi^T \xi \right]}, \quad (4)$$

where $\mathbf{t} = (t_1, t_2)^T$, $\xi = (\xi_1, \xi_2)^T$.

3. Main results

In this section, we propose a new definition of the PLCRT and derive several of its fundamental properties, convolution, correlation, Heisenberg's uncertainty principle.

3.1. Polar linear canonical ridgelet transform (PLCRT)

By overcoming the limitations of both the Radon transform, PWT and the LCT, the PLCRT provides a more flexible and comprehensive mathematical framework. Its enhanced degrees of freedom make it particularly effective in fields such as signal processing, and spectral analysis, especially in scenarios where classical transforms encounter challenges, this work lays a solid foundation for further research in these areas.

Definition 4. For any $f(\mathbf{t}), \psi(\mathbf{t}) \in L^2(\mathbb{R}^2)$, the matrix $M = (A, B; C, D)$, $AD - BC = 1$, the PLCRT is defined as

$$\mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f](a, \mathbf{b}) = \int_{\mathbb{R}^2} f(\mathbf{t}) \overline{\psi_{\mathbf{b}, a, \alpha, \gamma, \beta}^M(\mathbf{t})} d\mathbf{t} = \left\langle f, \psi_{\mathbf{b}, a, \alpha, \gamma, \beta}^M \right\rangle_{L^2(\mathbb{R}^2)}, \quad (5)$$

where the transform kernel $\overline{\psi_{\mathbf{b}, a, \alpha, \gamma, \beta}^M(\mathbf{t})}$ satisfies

$$\psi_{\mathbf{b}, a, \alpha, \gamma, \beta}^M(\mathbf{t}) = \frac{1}{a} \psi \left(\frac{R_\alpha(\Lambda_{\gamma, \beta} \mathbf{t} - \mathbf{b})}{a} \right) e^{j \left(\frac{A}{2B} \mathbf{b}^T \mathbf{b} - \frac{A}{2B} (\Lambda_{\gamma, \beta} \mathbf{t})^T (\Lambda_{\gamma, \beta} \mathbf{t}) \right)},$$

where $a \in \mathbb{R}^+$, $\mathbf{b}, \mathbf{t} \in \mathbb{R}^2$, $R_\alpha = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, $\Lambda_{\gamma, \beta} = \begin{pmatrix} \cos \gamma & \sin \gamma \\ \cos \beta & \sin \beta \end{pmatrix}$, $\sin(\beta - \gamma) = 1$, $\alpha, \beta, \gamma \in [0, 2\pi)$.

From the Definition 4, it can be seen that for the matrix $M = (A, B; C, D)$, the flexibility of the angles α, γ, β makes the localization characteristics of the PLCRT adjustable. Therefore, the PLCRT plays a crucial role in optimizing the solution of grid transforms and can effectively detect linear singularities in 2D signals such as images.

Property 1 (Linearity). For $f_1(\mathbf{t}), f_2(\mathbf{t}) \in L^2(\mathbb{R}^2)$, and $c_1, c_2 \in \mathbb{C}$, let $g(\mathbf{t}) = c_1 f_1(\mathbf{t}) + c_2 f_2(\mathbf{t})$, the PLCRT is linear, then

$$\mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[g(\mathbf{t})](a, \mathbf{b}) = c_1 \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f_1(\mathbf{t})](a, \mathbf{b}) + c_2 \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f_2(\mathbf{t})](a, \mathbf{b}). \quad (6)$$

The proof process is obviously easy, so it is omitted here.

Property 2 (Scaling). For $f(\mathbf{t}) \in L^2(\mathbb{R}^2)$, let $\mathbf{u} = s\mathbf{t}$, and scaling factor $s > 0$, the PLCRT satisfies the scaling property, then

$$\begin{aligned} & \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f(s\mathbf{t})](a, \mathbf{b}) \\ &= \frac{1}{s} e^{j \frac{A}{2B} ((\Lambda_{\gamma, \beta} \cdot \mathbf{u})^T (\Lambda_{\gamma, \beta} \cdot \mathbf{u}) \left(\frac{1}{s^2} - 1 \right) - \mathbf{b}^T \mathbf{b})} \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f(\mathbf{u})](as, s\mathbf{b}). \end{aligned} \quad (7)$$

Proof. Starting with the left-hand side, we have

$$\begin{aligned} & \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f(s\mathbf{t})](a, \mathbf{b}) \\ &= \frac{1}{a} \int_{\mathbb{R}^2} f(s\mathbf{t}) e^{j \frac{A}{2B} ((\Lambda_{\gamma, \beta} \cdot \mathbf{t})^T (\Lambda_{\gamma, \beta} \cdot \mathbf{t}) - \mathbf{b}^T \mathbf{b})} \overline{\psi \left(\frac{R_\alpha(\Lambda_{\gamma, \beta} \cdot \mathbf{t} - \mathbf{b})}{a} \right)} d\mathbf{t}, \end{aligned}$$

perform the change of variables $\mathbf{u} = \mathbf{s}\mathbf{t}$, so $\mathbf{t} = \frac{\mathbf{u}}{\mathbf{s}}$ and $d\mathbf{t} = \frac{1}{s^2}d\mathbf{u}$, we have

$$\begin{aligned} \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f(\mathbf{u})](a, \mathbf{b}) &= \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{u}) e^{j\frac{A}{2B} \left((\Lambda_{\gamma,\beta} \cdot \frac{\mathbf{u}}{s})^T (\Lambda_{\gamma,\beta} \cdot \frac{\mathbf{u}}{s}) - \mathbf{b}^T \mathbf{b} \right)} \overline{\psi \left(\frac{R_\alpha \left(\Lambda_{\gamma,\beta} \cdot \frac{\mathbf{u}}{s} - \mathbf{b} \right)}{a} \right)} \frac{1}{s^2} d\mathbf{u} \\ &= \frac{1}{as^2} \int_{\mathbb{R}^2} f(\mathbf{u}) e^{j\frac{A}{2B} \left(\frac{1}{s^2} (\Lambda_{\gamma,\beta} \cdot \mathbf{u})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{u}) - \mathbf{b}^T \mathbf{b} \right)} \overline{\psi \left(\frac{R_\alpha \left(\frac{1}{s} \Lambda_{\gamma,\beta} \cdot \mathbf{u} - \mathbf{b} \right)}{a} \right)} d\mathbf{u}, \end{aligned}$$

simplify the exponential term, we have

$$\begin{aligned} &e^{j\frac{A}{2B} \left(\frac{1}{s^2} (\Lambda_{\gamma,\beta} \cdot \mathbf{u})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{u}) - \mathbf{b}^T \mathbf{b} \right)} \\ &= e^{j\frac{A}{2B} \left(\frac{1}{s^2} (\Lambda_{\gamma,\beta} \cdot \mathbf{u})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{u}) - \mathbf{b}^T \mathbf{b} \right)} e^{-j\frac{A}{2B} \left((\Lambda_{\gamma,\beta} \cdot \mathbf{u})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{u}) - (\mathbf{s}\mathbf{b})^T (\mathbf{s}\mathbf{b}) \right)} \\ &\times e^{j\frac{A}{2B} \left((\Lambda_{\gamma,\beta} \cdot \mathbf{u})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{u}) - (\mathbf{s}\mathbf{b})^T (\mathbf{s}\mathbf{b}) \right)} \\ &= e^{j\frac{A}{2B} \left((\Lambda_{\gamma,\beta} \cdot \mathbf{u})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{u}) \left(\frac{1}{s^2} - 1 \right) - \mathbf{b}^T \mathbf{b} \right)} e^{j\frac{A}{2B} \left((\Lambda_{\gamma,\beta} \cdot \mathbf{u})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{u}) - (\mathbf{s}\mathbf{b})^T (\mathbf{s}\mathbf{b}) \right)}, \end{aligned}$$

putting it all together, we have

$$\begin{aligned} \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f(\mathbf{u})](a, \mathbf{b}) &= \frac{1}{as^2} \int_{\mathbb{R}^2} f(\mathbf{u}) e^{j\frac{A}{2B} \left(\frac{1}{s^2} (\Lambda_{\gamma,\beta} \cdot \mathbf{u})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{u}) - \mathbf{b}^T \mathbf{b} \right)} \overline{\psi \left(\frac{R_\alpha \left(\Lambda_{\gamma,\beta} \cdot \mathbf{u} - \mathbf{s}\mathbf{b} \right)}{as} \right)} d\mathbf{u} \\ &= \frac{1}{s} e^{j\frac{A}{2B} \left((\Lambda_{\gamma,\beta} \cdot \mathbf{u})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{u}) \left(\frac{1}{s^2} - 1 \right) - \mathbf{b}^T \mathbf{b} \right)} \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f(\mathbf{u})](as, \mathbf{s}\mathbf{b}). \end{aligned}$$

The proof is completed.

Property 3 (Translation). For $f(\mathbf{t}) \in L^2(\mathbb{R}^2)$, $h(\mathbf{t}) = f(\mathbf{t} - \mathbf{t}_0) e^{-j\frac{A}{B} \mathbf{t}^T \mathbf{t}_0}$, then

$$\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[h(\mathbf{t})](a, \mathbf{b}) = \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f(\mathbf{t})](a, \mathbf{b} - \Lambda_{\gamma,\beta} \cdot \mathbf{t}_0) e^{j\frac{A}{B} (\Lambda_{\gamma,\beta} \cdot \mathbf{b})^T \mathbf{t}_0}. \quad (8)$$

Proof. By definition, the PLCRT of $h(\mathbf{t})$ is given by

$$\begin{aligned} \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[h(\mathbf{t})](a, \mathbf{b}) &= \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{t} - \mathbf{t}_0) e^{-j\frac{A}{B} \mathbf{t}^T \mathbf{t}_0} e^{j\frac{A}{2B} \left((\Lambda_{\gamma,\beta} \cdot \mathbf{x})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{t}) - \mathbf{b}^T \mathbf{b} \right)} \overline{\psi \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \cdot \mathbf{t} - \mathbf{b})}{a} \right)} d\mathbf{t}, \end{aligned}$$

perform the change of variables $\mathbf{y} = \mathbf{t} - \mathbf{t}_0$, we have

$$\begin{aligned} \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f(\mathbf{y})](a, \mathbf{b}) &= \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{y}) e^{-j\frac{A}{B} (\mathbf{y} + \mathbf{t}_0)^T \mathbf{t}_0} e^{j\frac{A}{2B} \left((\Lambda_{\gamma,\beta} \cdot (\mathbf{y} + \mathbf{t}_0))^T (\Lambda_{\gamma,\beta} \cdot (\mathbf{y} + \mathbf{t}_0)) - \mathbf{b}^T \mathbf{b} \right)} \\ &\times \overline{\psi \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \cdot (\mathbf{y} + \mathbf{t}_0) - \mathbf{b})}{a} \right)} d\mathbf{y} \\ &= \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{y}) e^{j\frac{A}{2B} \left((\Lambda_{\gamma,\beta} \cdot \mathbf{y})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{y}) - \mathbf{b}^T \mathbf{b} \right)} e^{j\frac{A}{B} (\Lambda_{\gamma,\beta} \cdot \mathbf{y})^T (\Lambda_{\gamma,\beta} \cdot \mathbf{t}_0)} \\ &\times \overline{\psi \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \cdot \mathbf{y} - (\mathbf{b} - \Lambda_{\gamma,\beta} \cdot \mathbf{t}_0))}{a} \right)} d\mathbf{y} \\ &= \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f(\mathbf{t})](a, \mathbf{b} - \Lambda_{\gamma,\beta} \cdot \mathbf{t}_0) e^{j\frac{A}{B} (\Lambda_{\gamma,\beta} \cdot \mathbf{b})^T \mathbf{t}_0}. \end{aligned}$$

The proof is completed.

Property 4. For any $f(\mathbf{t}) \in L^2(\mathbb{R}^2)$, the PLCRT can be expressed via the 2D-LCT as

$$L^M \left(\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b}) \right) (\xi)$$

$$= \frac{a}{2\pi B} e^{j\left(\frac{D}{2B} \xi^T \xi\right)} \mathcal{F} [G^M] \left(\frac{\Lambda_{\gamma,\beta} \cdot \xi}{B} \right) \mathcal{F} [\psi] \left(\frac{a R_\alpha^{-1} \cdot \xi}{B} \right), \quad (9)$$

where $\mathcal{F}[\psi]$ is the 2D Fourier transform of the wavelet ψ .

Proof. To establish the relationship between the PLCRT and the LCT, we will proceed through the following

$$\begin{aligned} L^M \left(\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b}) \right) (\xi) &= \frac{1}{2\pi B} \int_{\mathbb{R}^2} \left[\int_{\mathbb{R}^2} f(\mathbf{t}) \overline{\psi_{a,\mathbf{b},\alpha,\gamma,\beta}^M(\mathbf{t})} d\mathbf{t} \right] \cdot e^{j\left[\frac{A}{2B} \mathbf{b}^T \mathbf{b} - \frac{1}{B} \mathbf{b}^T \cdot \xi + \frac{D}{2B} \xi^T \xi\right]} d\mathbf{b} \\ &= \frac{1}{2\pi a B} \int_{\mathbb{R}^2} \left[\int_{\mathbb{R}^2} f(\mathbf{t}) \overline{\psi \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \mathbf{t} - \mathbf{b})}{a} \right)} e^{-j\left(\frac{A}{2B} \mathbf{b}^T \mathbf{b} - \frac{A}{2B} (\Lambda_{\gamma,\beta} \mathbf{t})^T (\Lambda_{\gamma,\beta} \mathbf{t})\right)} d\mathbf{t} \right] \\ &\times e^{j\left[\frac{A}{2B} \mathbf{b}^T \mathbf{b} - \frac{1}{B} \mathbf{b}^T \cdot \xi + \frac{D}{2B} \xi^T \xi\right]} d\mathbf{b} \\ &= \frac{1}{2\pi a B} \int_{\mathbb{R}^2} \left[\int_{\mathbb{R}^2} f(\mathbf{t}) e^{j\left[\frac{A}{2B} (\Lambda_{\gamma,\beta} \mathbf{t})^T (\Lambda_{\gamma,\beta} \mathbf{t})\right]} d\mathbf{t} \right] \overline{\psi \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \mathbf{t} - \mathbf{b})}{a} \right)} \\ &\times e^{j\left[-\frac{1}{B} \mathbf{b}^T \cdot \xi + \frac{D}{2B} \xi^T \xi\right]} d\mathbf{b}, \end{aligned}$$

next, for the convenience of writing, let the $G^M(\mathbf{t}) = f(\mathbf{t}) e^{j\left[\frac{A}{2B} (\Lambda_{\gamma,\beta} \mathbf{t})^T (\Lambda_{\gamma,\beta} \mathbf{t})\right]}$, perform a change of variables, let the $\mathbf{u} = \frac{R_\alpha (\Lambda_{\gamma,\beta} \mathbf{t} - \mathbf{b})}{a}$, so $\mathbf{b} = \Lambda_{\gamma,\beta} \cdot \mathbf{t} - a R_\alpha^{-1} \cdot \mathbf{u}$, $(R_\alpha)^{-1} = (R_\alpha)^T$, and $d\mathbf{b} = \det(-a R_\alpha^{-1}) d\mathbf{u} = a^2 d\mathbf{u}$, we have

$$\begin{aligned} &\frac{1}{2\pi a B} \int_{\mathbb{R}^2} \left[\int_{\mathbb{R}^2} f(\mathbf{t}) e^{j\left[\frac{A}{2B} (\Lambda_{\gamma,\beta} \mathbf{t})^T (\Lambda_{\gamma,\beta} \mathbf{t})\right]} d\mathbf{t} \right] \overline{\psi \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \mathbf{t} - \mathbf{b})}{a} \right)} \\ &\times e^{j\left[-\frac{1}{B} \mathbf{b}^T \cdot \xi + \frac{D}{2B} \xi^T \xi\right]} d\mathbf{b} \\ &= \frac{a}{2\pi B} e^{j\left(\frac{D}{2B} \xi^T \xi\right)} \int_{\mathbb{R}^2} G^M(\mathbf{t}) e^{-\frac{j}{B} (\Lambda_{\gamma,\beta} \mathbf{t})^T \cdot \xi} d\mathbf{t} \int_{\mathbb{R}^2} \overline{\psi(\mathbf{u})} e^{-\frac{j}{B} (a R_\alpha^{-1} \cdot \mathbf{u})^T \cdot \xi} d\mathbf{u} \\ &= \frac{a}{2\pi B} e^{j\left(\frac{D}{2B} \xi^T \xi\right)} \mathcal{F} [G^M] \left(\frac{\Lambda_{\gamma,\beta} \cdot \xi}{B} \right) \overline{\mathcal{F} [\psi] \left(\frac{a R_\alpha^{-1} \cdot \xi}{B} \right)}. \end{aligned}$$

The proof is completed.

Theorem 1. For any wavelet $\psi(\mathbf{t})$ and $\phi(\mathbf{t})$, $f(\mathbf{t})$ and $g(\mathbf{t}) \in L^2(\mathbb{R}^2)$ satisfy the following orthogonality relation

$$\begin{aligned} &\int_{\alpha,\beta,\gamma} \int_{\mathbb{R}^2} \int_{\mathbb{R}^+} \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f(\mathbf{t})](a, \mathbf{b}) \overline{\mathcal{R}_{\alpha,\gamma,\beta}^{M,\phi}[g(\mathbf{t})](a, \mathbf{b})} \frac{da d\alpha d\gamma d\beta d\mathbf{b}}{a^3} \\ &= C_{\psi,\phi} \langle f, g \rangle_2, \end{aligned} \quad (10)$$

where $\int_{\alpha,\beta,\gamma} = \int_0^{2\pi} \int_0^{2\pi} \int_0^{2\pi}$, $C_{\psi,\phi}$ is given by the usual cross-admissibility condition, then

$$C_{\psi,\phi} = \int_{\mathbb{R}^2} \frac{\overline{\mathcal{F}[\psi](\boldsymbol{\omega})} \cdot \mathcal{F}[\phi](\boldsymbol{\omega})}{\boldsymbol{\omega}^2} d\boldsymbol{\omega} < \infty, \quad (11)$$

where $\boldsymbol{\omega}^2 = |\boldsymbol{\omega}|^2 = \omega_1^2 + \omega_2^2$.

Proof. Applying the Parseval's formula for the linear canonical transform together with a change in the order of integration, we have

$$\begin{aligned} &\int_{\alpha,\beta,\gamma} \int_{\mathbb{R}^2} \int_{\mathbb{R}^+} \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b}) \overline{\mathcal{R}_{\alpha,\gamma,\beta}^{M,\phi}[g](a, \mathbf{b})} \frac{da d\alpha d\gamma d\beta d\mathbf{b}}{a^3} \\ &= \int_{\alpha,\beta,\gamma} \int_{\mathbb{R}^2} \int_{\mathbb{R}^+} L^M \left(\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b}) \right) (\xi) \cdot \overline{L^M \left(\mathcal{R}_{\alpha,\gamma,\beta}^{M,\phi}[g](a, \mathbf{b}) \right) (\xi)} \end{aligned}$$

$$\begin{aligned}
& \times \frac{da d\alpha d\gamma d\beta d\xi}{a^3} \\
& = \int_{\alpha, \beta, \gamma} \int_{\mathbb{R}^2} \int_{\mathbb{R}^+} \frac{a}{2\pi B} e^{j\left(\frac{D}{2B} \xi^T \xi\right)} \mathcal{F}[G^M] \left(\frac{\Lambda_{\gamma, \beta} \cdot \xi}{B} \right) \overline{\mathcal{F}[\psi] \left(\frac{aR_{\alpha}^{-1} \cdot \xi}{B} \right)} \\
& \times \frac{a}{2\pi B} e^{j\left(\frac{D}{2B} \xi^T \xi\right)} \mathcal{F}[H^M] \left(\frac{\Lambda_{\gamma, \beta} \cdot \xi}{B} \right) \overline{\mathcal{F}[\phi] \left(\frac{aR_{\alpha}^{-1} \cdot \xi}{B} \right)} \frac{da d\alpha d\gamma d\beta d\xi}{a^3} \\
& = \int_{\beta, \gamma} \int_{\mathbb{R}^2} \mathcal{F}[G^M] \left(\frac{\Lambda_{\gamma, \beta} \cdot \xi}{B} \right) \overline{\mathcal{F}[H^M] \left(\frac{\Lambda_{\gamma, \beta} \cdot \xi}{B} \right)} d\gamma d\beta d\xi \\
& \times \int_{\alpha} \int_{\mathbb{R}^+} \left(\frac{a}{2\pi B} \right)^2 \mathcal{F}[\phi] \left(\frac{aR_{\alpha}^{-1} \cdot \xi}{B} \right) \overline{\mathcal{F}[\psi] \left(\frac{aR_{\alpha}^{-1} \cdot \xi}{B} \right)} \frac{da d\alpha}{a^3},
\end{aligned}$$

where the functions $G^M(\mathbf{t})$ and $H^M(\mathbf{t})$ are given by

$$G^M(\mathbf{t}) = f(\mathbf{t}) e^{j\left[\frac{A}{2B} (\Lambda_{\gamma, \beta} \mathbf{t})^T (\Lambda_{\gamma, \beta} \mathbf{t})\right]},$$

and

$$H^M(\mathbf{t}) = g(\mathbf{t}) e^{j\left[\frac{A}{2B} (\Lambda_{\gamma, \beta} \mathbf{t})^T (\Lambda_{\gamma, \beta} \mathbf{t})\right]}.$$

Next, let the $\frac{aR_{\alpha}^{-1} \cdot \xi}{B} = \omega$, $dad\alpha = \frac{B^2}{a|\xi|^2} d\omega$, $a^2 = B^2 \frac{|\omega|^2}{|\xi|^2}$, then

$$\begin{aligned}
& \left(\frac{a}{2\pi B} \right)^2 \int_0^{2\pi} \int_{\mathbb{R}^+} \mathcal{F}[\psi] \left(\frac{aR_{\alpha}^{-1} \cdot \xi}{B} \right) \cdot \mathcal{F}[\phi] \left(\frac{aR_{\alpha}^{-1} \cdot \xi}{B} \right) \frac{dad\alpha}{a^3} \\
& = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \overline{\mathcal{F}[\psi](\omega)} \cdot \mathcal{F}[\phi](\omega) \frac{dad\alpha}{a^2} \\
& = \frac{1}{B^2} \frac{1}{(2\pi)^2} C_{\psi, \phi}.
\end{aligned}$$

Let $\frac{\Lambda_{\gamma, \beta} \cdot \xi}{B} = \zeta$, $d\xi = \frac{B^2}{|\sin(\beta - \gamma)|} d\zeta$, invoking the cross-admissibility condition, we have

$$\begin{aligned}
& \int_{\alpha, \beta, \gamma} \int_{\mathbb{R}^2} \int_{\mathbb{R}^+} \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f](a, \mathbf{b}) \overline{\mathcal{R}_{\alpha, \gamma, \beta}^{M, \phi}[g](a, \mathbf{b})} \frac{da d\alpha d\gamma d\beta d\mathbf{b}}{a^3} \\
& = \frac{1}{B^2} \frac{1}{(2\pi)^2} C_{\psi, \phi} \int_{\mathbb{R}^2} \int_{\beta, \gamma} \mathcal{F}[G^M] \left(\frac{\Lambda_{\gamma, \beta} \cdot \xi}{B} \right) \overline{\mathcal{F}[H^M] \left(\frac{\Lambda_{\gamma, \beta} \cdot \xi}{B} \right)} d\gamma d\beta d\xi \\
& = \frac{1}{B^2} \frac{1}{(2\pi)^2} C_{\psi, \phi} \int_{\mathbb{R}^2} \int_{\beta, \gamma} \mathcal{F}[G^M](\zeta) \overline{\mathcal{F}[H^M](\zeta)} \frac{B^2}{|\sin(\beta - \gamma)|} d\gamma d\beta d\zeta \\
& = C_{\psi, \phi} \int_{\mathbb{R}^2} \mathcal{F}[G^M](\zeta) \overline{\mathcal{F}[H^M](\zeta)} d\zeta \\
& = C_{\psi, \phi} \int_{\mathbb{R}^2} G^M(\mathbf{t}) \overline{H^M(\mathbf{t})} d\mathbf{t} \\
& = C_{\psi, \phi} \int_{\mathbb{R}^2} f(\mathbf{t}) \overline{g(\mathbf{t})} d\mathbf{t} \\
& = C_{\psi, \phi} \langle f, g \rangle_2.
\end{aligned}$$

The proof is completed.

Property 5 (Inversion formula). For any $f(\mathbf{t}) \in L^2(\mathbb{R}^2)$ can be reconstructed from the corresponding $\mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f](a, \mathbf{b})$ via the following formula

$$f(\mathbf{t}) = \frac{1}{C_{\psi, \phi}} \int_{\alpha, \gamma, \beta} \int_{\mathbb{R}^2} \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f](a, \mathbf{b}) \phi_{a, \mathbf{b}, \alpha, \gamma, \beta}^M(\mathbf{t}) \frac{da d\alpha d\gamma d\beta d\mathbf{b}}{a^3}, \quad (12)$$

where $C_{\psi, \phi}$ is given by the cross-admissibility condition in the Theorem 1.

Proof. By Theorem 1, we choose $g(\mathbf{x}) = \delta(\mathbf{x} - \mathbf{t})$, $\mathbf{t} \in \mathbb{R}^2$, so that

$$\mathcal{R}_{\alpha, \gamma, \beta}^{M, \phi}[g(\mathbf{x})](a, \mathbf{b}) = \overline{\phi_{a, \mathbf{b}, \alpha, \gamma, \beta}^M(\mathbf{t})},$$

in addition, we note that

$$\langle f, g \rangle_2 = \int_{\mathbb{R}^2} f(\mathbf{x}) \delta(\mathbf{x} - \mathbf{t}) d\mathbf{t} = f(\mathbf{t}),$$

next, applying the Theorem 1, we have

$$f(\mathbf{t}) = \frac{1}{C_{\psi, \phi}} \int_{\alpha, \gamma, \beta} \int_{\mathbb{R}^2} \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f](a, \mathbf{b}) \phi_{a, \mathbf{b}, \alpha, \gamma, \beta}^M(\mathbf{t}) \frac{da d\alpha d\gamma d\beta d\mathbf{b}}{a^3}.$$

The proof is completed.

3.2. Convolution theorem and correlation theorem

Theorem 2 (Convolution theorem). For ψ_f and ψ_g are two admissible wavelets, if $h = (f * g)$ and $\psi_h = (\psi_f * \psi_g)$, then

$$\mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[h(\mathbf{x})](a, \mathbf{b}) = \left[\mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[f] * \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[g] \right](a, \mathbf{b}). \quad (13)$$

Proof. According to the classical theory of convolution, the wavelet convolution

$$\psi_h = (\psi_f * \psi_g)(\mathbf{t}) = \int_{\mathbb{R}^2} \psi_f(R_{\alpha} \mathbf{y}) \psi_g \left(\frac{R_{\alpha} (\Lambda_{\gamma, \beta} \cdot \mathbf{x} - \mathbf{b} - a\mathbf{y})}{a} \right) d\mathbf{y}. \quad (14)$$

Next, substitute the convolution of $h(\mathbf{x})$, justified by Fubini's theorem, as all functions involved are integrable, interchange the order of integration, we have

$$\begin{aligned}
& \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[h(\mathbf{x})](a, \mathbf{b}) \\
& = \frac{1}{a} \int_{\mathbb{R}^2} h(\mathbf{x}) \psi_h \left(\frac{R_{\alpha} (\Lambda_{\gamma, \beta} \cdot \mathbf{x} - \mathbf{b})}{a} \right) e^{j\frac{A}{2B} ((\Lambda_{\gamma, \beta} \cdot \mathbf{x})^T (\Lambda_{\gamma, \beta} \cdot \mathbf{x}) - \mathbf{b}^T \mathbf{b})} d\mathbf{x} \\
& = \frac{1}{a} e^{j\frac{A}{2B} ((\Lambda_{\gamma, \beta} \cdot \mathbf{x})^T (\Lambda_{\gamma, \beta} \cdot \mathbf{x}) - \mathbf{b}^T \mathbf{b})} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \left(\int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{x} - \mathbf{t}) \right) \overline{\psi_f(R_{\alpha} \mathbf{y})} \\
& \times \psi_g \left(\frac{R_{\alpha} (\Lambda_{\gamma, \beta} \cdot \mathbf{x} - \mathbf{b} - a\mathbf{y})}{a} \right) d\mathbf{t} d\mathbf{x} d\mathbf{y}.
\end{aligned}$$

Next, let $\mathbf{m} = \mathbf{x} - \mathbf{t}$, so $\mathbf{n} = \mathbf{b} + a\mathbf{y} - \Lambda_{\gamma, \beta} \mathbf{t}$ satisfy $2\mathbf{n}^T \mathbf{n} - 2\mathbf{b}^T \mathbf{n} + 2\Lambda_{\gamma, \beta} \mathbf{m}^T \mathbf{t} = 0$, we have

$$\begin{aligned}
& \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi}[h(\mathbf{x})](a, \mathbf{b}) \\
& = \frac{1}{a^2} e^{j\frac{A}{2B} [(\Lambda_{\gamma, \beta} \cdot \mathbf{t})^T (\Lambda_{\gamma, \beta} \cdot \mathbf{t}) - (\mathbf{b} - \mathbf{n})^T (\mathbf{b} - \mathbf{n}) + (\Lambda_{\gamma, \beta} \cdot \mathbf{m})^T (\Lambda_{\gamma, \beta} \cdot \mathbf{m}) - \mathbf{n}^T \mathbf{n}]} \\
& \times \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{m}) \\
& \times \overline{\psi_f \left(\frac{R_{\alpha} (\Lambda_{\gamma, \beta} \cdot \mathbf{t} - (\mathbf{b} - \mathbf{n}))}{a} \right)} \psi_g \left(\frac{R_{\alpha} (\Lambda_{\gamma, \beta} \cdot \mathbf{m} - \mathbf{n})}{a} \right) d\mathbf{t} d\mathbf{n} d\mathbf{m} \\
& = \int_{\mathbb{R}^2} \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{t}) e^{j\frac{A}{2B} [(\Lambda_{\gamma, \beta} \cdot \mathbf{t})^T (\Lambda_{\gamma, \beta} \cdot \mathbf{t}) - (\mathbf{b} - \mathbf{n})^T (\mathbf{b} - \mathbf{n})]} \\
& \times \overline{\psi_f \left(\frac{R_{\alpha} (\Lambda_{\gamma, \beta} \cdot \mathbf{t} - (\mathbf{b} - \mathbf{n}))}{a} \right)} d\mathbf{t} \\
& \times \frac{1}{a} \int_{\mathbb{R}^2} g(\mathbf{m}) e^{j\frac{A}{2B} [(\Lambda_{\gamma, \beta} \cdot \mathbf{m})^T (\Lambda_{\gamma, \beta} \cdot \mathbf{m}) - \mathbf{n}^T \mathbf{n}]} \psi_g \left(\frac{R_{\alpha} (\mathbf{m} - \mathbf{n})}{a} \right) d\mathbf{m} d\mathbf{n}
\end{aligned}$$

$$= \int_{\mathbb{R}^2} \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b} - \mathbf{n}) \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[g](a, \mathbf{n}) d\mathbf{n}$$

$$= [\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f] * \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[g]](a, \mathbf{b}).$$

The proof is completed.

Theorem 3 (Correlation theorem). For ψ_f and ψ_g are two admissible wavelets, if $h = (f * g)$ and $\psi_h = (\psi_f * \psi_g)$, then

$$\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[h(\mathbf{x})](a, \mathbf{b}) = [\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f] \odot \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[g]](a, -\mathbf{b}). \quad (15)$$

Proof. According to the classical theory of correlation, the wavelet correlation

$$(\psi_f \odot \psi_g)(\mathbf{t}) = \int_{\mathbb{R}^2} \psi_f(R_\alpha \mathbf{y}) \psi_g \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \cdot \mathbf{x} - \mathbf{b} + a\mathbf{y})}{a} \right) d\mathbf{y}. \quad (16)$$

By the definition equation (16) of the correlation operators and Fubini's theorem, as all functions involved are integrable, interchange the order of integration, we have

$$\begin{aligned} & \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[h(\mathbf{x})](a, \mathbf{b}) \\ &= \frac{1}{a} e^{j\frac{A}{2B}((\Lambda_{\gamma,\beta} \cdot \mathbf{x})^2 - \mathbf{b}^2)} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \left(\int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{x} + \mathbf{t}) \right) \overline{\psi_f(R_\alpha \mathbf{y})} \\ & \quad \times \psi_g \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \cdot \mathbf{x} - \mathbf{b} + a\mathbf{y})}{a} \right) d\mathbf{t} dx dy. \end{aligned}$$

Next, let $\mathbf{m} = \mathbf{x} + \mathbf{t}$, so $\mathbf{n} = \mathbf{b} - a\mathbf{y} + \Lambda_{\gamma,\beta} \mathbf{t}$ satisfy $2\mathbf{n}^\top \mathbf{n} - 2\mathbf{b}^\top \mathbf{n} + 2\Lambda_{\gamma,\beta} \mathbf{m}^\top \mathbf{t} = 0$, we have

$$\begin{aligned} & \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[h(\mathbf{x})](a, \mathbf{b}) \\ &= \frac{1}{a^2} e^{j\frac{A}{2B}[(\Lambda_{\gamma,\beta} \cdot \mathbf{t})^\top (\Lambda_{\gamma,\beta} \cdot \mathbf{t}) - (\mathbf{n} - \mathbf{b})^\top (\mathbf{n} - \mathbf{b}) + (\Lambda_{\gamma,\beta} \cdot \mathbf{m})^\top (\Lambda_{\gamma,\beta} \cdot \mathbf{m}) - \mathbf{n}^\top \mathbf{n}]} \\ & \quad \times \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} f(\mathbf{t}) g(\mathbf{m}) \\ & \quad \times \overline{\psi_f \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \cdot \mathbf{t} - (\mathbf{n} - \mathbf{b}))}{a} \right)} \psi_g \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \cdot \mathbf{m} - \mathbf{n})}{a} \right) d\mathbf{t} d\mathbf{n} d\mathbf{m} \\ &= \int_{\mathbb{R}^2} \frac{1}{a} \int_{\mathbb{R}^2} f(\mathbf{t}) e^{j\frac{A}{2B}[(\Lambda_{\gamma,\beta} \cdot \mathbf{t})^\top (\Lambda_{\gamma,\beta} \cdot \mathbf{t}) - (\mathbf{n} - \mathbf{b})^\top (\mathbf{n} - \mathbf{b})]} \\ & \quad \times \overline{\psi_f \left(\frac{R_\alpha (\Lambda_{\gamma,\beta} \cdot \mathbf{t} - (\mathbf{n} - \mathbf{b}))}{a} \right)} d\mathbf{t} \\ & \quad \times \frac{1}{a} \int_{\mathbb{R}^2} g(\mathbf{m}) e^{j\frac{A}{2B}[(\Lambda_{\gamma,\beta} \cdot \mathbf{m})^\top (\Lambda_{\gamma,\beta} \cdot \mathbf{m}) - \mathbf{n}^\top \mathbf{n}]} \overline{\psi_g \left(\frac{R_\alpha (\mathbf{m} - \mathbf{n})}{a} \right)} d\mathbf{m} d\mathbf{n} \\ &= \int_{\mathbb{R}^2} \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{n} - \mathbf{b}) \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[g](a, \mathbf{n}) d\mathbf{n} \\ &= [\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f] \odot \mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[g]](a, -\mathbf{b}). \end{aligned}$$

The proof is completed.

3.3. Uncertainty principles

One of the key discoveries in modern physics is the uncertainty principle, which plays a crucial role in explaining various phenomena across multiple scientific domains. Currently, numerous integral transforms have been extensively investigated, leading to significant theoretical advancements [17–19,31].

Theorem 4 (Heisenberg's inequality). For any $f \in L^2(\mathbb{R}^2)$, $\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b})$ is the PLCRT of f , then

$$\begin{aligned} & \left\{ \int_{\alpha,\beta,\gamma, \mathbb{R}^+} \int_{\mathbb{R}^2} \mathbf{b}^2 |\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b})|^2 \frac{da d\alpha d\gamma d\beta d\mathbf{b}}{a^3} \right\}^{1/2} \\ & \times \left\{ \int_{\mathbb{R}^2} \xi^2 |\mathcal{F}[G^M](\xi)|^2 d\xi \right\}^{1/2} \\ & \geq \frac{\sqrt{C_{\psi,\psi}}}{2} \|f\|_{L(\mathbb{R}^2)}^2. \end{aligned} \quad (17)$$

Proof. For any $g \in L^2(\mathbb{R}^2)$, Heisenberg's uncertainty principle associated with the LCT reads [31]:

$$\left\{ \int_{\mathbb{R}^2} \mathbf{t}^2 |g(\mathbf{t})|^2 d\mathbf{t} \right\}^{1/2} \left\{ \int_{\mathbb{R}^2} \xi^2 |L^M[g](\xi)|^2 d\xi \right\}^{1/2} \geq \frac{|B|}{2} \left\{ \int_{\mathbb{R}^2} |g(\mathbf{t})|^2 d\mathbf{t} \right\}, \quad (18)$$

we shall identify the PLCRT $\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b})$ as a function of the translation variable $\mathbf{b}$, then

$$\begin{aligned} & \int_{\alpha,\beta,\gamma, \mathbb{R}^+} \int_{\mathbb{R}^2} \mathbf{b}^2 |\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b})|^2 d\mathbf{b} \\ & \times \left\{ \int_{\mathbb{R}^2} \xi^2 |L^M[\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b})](\xi)|^2 d\xi \right\}^{1/2} \frac{da d\alpha d\gamma d\beta}{a^3} \\ & \geq \frac{|B|}{2} \left\{ \int_{\alpha,\beta,\gamma, \mathbb{R}^+} \int_{\mathbb{R}^2} |\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b})|^2 \frac{da d\alpha d\gamma d\beta d\mathbf{b}}{a^3} \right\}. \end{aligned} \quad (19)$$

Applying the Theorem 1, we have

$$\begin{aligned} & \left\{ \int_{\alpha,\beta,\gamma, \mathbb{R}^+} \int_{\mathbb{R}^2} \mathbf{b}^2 |\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b})|^2 \frac{da d\alpha d\gamma d\beta d\mathbf{b}}{a^3} \right\}^{1/2} \\ & \times \left\{ \int_{\alpha,\beta,\gamma, \mathbb{R}^+} \int_{\mathbb{R}^2} \xi^2 |L^M[\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b})](\xi)|^2 \frac{d\xi da d\alpha d\gamma d\beta}{a^3} \right\}^{1/2} \\ & \geq \frac{|B|}{2} C_{\psi,\psi} \langle f, f \rangle_2. \end{aligned} \quad (20)$$

Using Property 4 and the Theorem 1 admissibility condition, we have

$$\begin{aligned} & \int_{\alpha,\beta,\gamma, \mathbb{R}^+} \int_{\mathbb{R}^2} \xi^2 |L^M[\mathcal{R}_{\alpha,\gamma,\beta}^{M,\psi}[f](a, \mathbf{b})](\xi)|^2 \frac{d\xi da d\alpha d\gamma d\beta}{a^3} \\ &= \int_{\mathbb{R}^2} \int_{\alpha,\beta,\gamma, \mathbb{R}^+} \xi^2 \left| \frac{a}{2\pi B} e^{j(\frac{D}{2B}\xi^\top \xi)} \mathcal{F}[G^M] \left(\frac{\Lambda_{\gamma,\beta} \cdot \xi}{B} \right) \right|^2 \\ & \quad \times \frac{d\xi da d\alpha d\gamma d\beta}{a^3} \\ &= \frac{a^2}{(2\pi B)^2} \int_{\mathbb{R}^2} \int_{\beta,\gamma} \xi^2 \left| \mathcal{F}[G^M] \left(\frac{\Lambda_{\gamma,\beta} \cdot \xi}{B} \right) \right|^2 d\xi d\gamma d\beta \end{aligned} \quad (21)$$

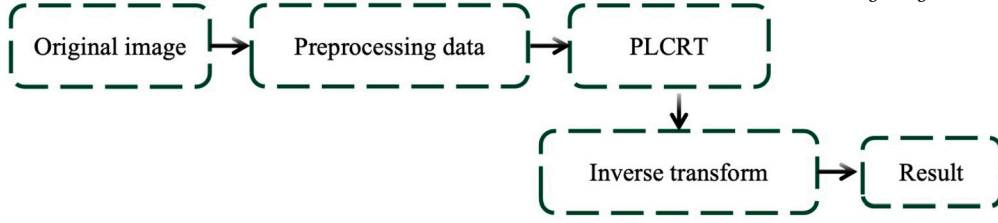


Fig. 1. The PLCRT retinal edge detection technology route.

$$\times \int_{\alpha} \int_{\mathbb{R}^+} \left| F[\psi] \left(\frac{aR_{\alpha}^{-1} \cdot \xi}{B} \right) \right|^2 \frac{da d\alpha}{a^3}.$$

Next, let the $\frac{aR_{\alpha}^{-1} \cdot \xi}{B} = \omega$, $da d\alpha = \frac{B^2}{a|\xi|^2} d\omega$, $a^2 = B^2 \frac{|\omega|^2}{|\xi|^2}$, we have

$$\frac{a^2}{(2\pi B)^2} \int_{\alpha} \int_{\mathbb{R}^+} \left| F[\psi] \left(\frac{aR_{\alpha}^{-1} \cdot \xi}{B} \right) \right|^2 \frac{da d\alpha}{a^3} = \frac{1}{(2\pi B)^2} C_{\psi, \psi}. \quad (22)$$

Similarly, let the $\frac{\Lambda_{\gamma, \beta} \cdot \xi}{B} = \zeta$, $d\xi = \frac{B^2}{|\sin(\beta - \gamma)|} d\zeta$, invoking the cross-admissibility condition, we have

$$\begin{aligned} & \int_{\alpha, \beta, \gamma} \int_{\mathbb{R}^+} \int_{\mathbb{R}^2} \xi^2 \left| L^M \left[\mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi} [f](a, \mathbf{b}) \right] (\xi) \right|^2 \frac{d\xi da d\alpha d\gamma d\beta}{a^3} \\ &= \frac{1}{B^2} \frac{1}{(2\pi)^2} C_{\psi, \phi} \int_{\mathbb{R}^2} \int_{\beta, \gamma} \xi^2 \left| F[G^M] \left(\frac{\Lambda_{\gamma, \beta} \cdot \xi}{B} \right) \right|^2 d\xi d\gamma d\beta \\ &= B^2 C_{\psi, \psi} \int_{\mathbb{R}^2} \xi^2 \left| F[G^M] (\zeta) \right|^2 d\zeta. \end{aligned} \quad (23)$$

Finally, we obtain the uncertainty inequality for the PLCRT as

$$\begin{aligned} & \left\{ \int_{\alpha, \beta, \gamma} \int_{\mathbb{R}^+} \int_{\mathbb{R}^2} \mathbf{b}^2 \left| \mathcal{R}_{\alpha, \gamma, \beta}^{M, \psi} [f](a, \mathbf{b}) \right|^2 \frac{da d\alpha d\gamma d\beta d\mathbf{b}}{a^3} \right\}^{1/2} \\ & \times \left\{ \int_{\mathbb{R}^2} \xi^2 \left| F[G^M] (\zeta) \right|^2 d\zeta \right\}^{1/2} \\ & \geq \frac{\sqrt{C_{\psi, \psi}}}{2} \|f\|_{L(\mathbb{R}^2)}^2. \end{aligned} \quad (24)$$

The proof is completed.

4. Potential application

In this section, we first selected diabetic retinopathy (DR) images from the STARE database [32] and applied the PLCRT algorithm for edge detection, quantifying its performance by computing mean squared error (MSE) and peak signal to noise ratio (PSNR) against expert annotated ground truth. To comprehensively validate the PLCRT's superiority, we conducted a systematic comparison with classical detectors (Sobel [33], Roberts [34], LoG [35], Prewitt [36], Canny [37]), the FOMSK operator of J. E. Lavín-Delgado [38], and the QRFODD operator of Kanwarpreet Kaur [39], while also evaluating the PLCRT's stability under salt-and-pepper and Gaussian noise; concurrently, we approximated each algorithm's time complexity and empirical runtime. Finally, we extended the PLCRT to clinical retinal images acquired in a hospital setting. Experimental results demonstrate that the PLCRT not only accurately extracts fine edge details in pathological regions but also exhibits

Table 1

Performance metrics for Background DR edge detection.

Retinopathy	Operator	MSE	PSNR
Background DR	Sobel	12810.2607	7.0552
	Roberts	12811.1172	7.0549
	LoG	12801.0477	7.0583
	Prewitt	12810.3090	7.0552
	Canny	12793.6126	7.0608
	FOMSK	10755.3754	7.8145
	QRFODD	9579.9030	8.3172
	PLCRT	8157.1510	9.0154

Table 2

Performance metrics for Stellate Maculopathy edge detection.

Retinopathy	Operator	MSE	PSNR
Stellate Maculopathy	Sobel	17020.2497	5.8211
	Roberts	17021.6201	5.8207
	LoG	17008.9057	5.8240
	Prewitt	17020.3213	5.8211
	Canny	16996.4967	5.8272
	FOMSK	13412.3240	6.8557
	QRFODD	13259.5560	6.9055
	PLCRT	8745.3681	8.7130

Table 3

Performance metrics for Proliferative DR edge detection.

Retinopathy	Operator	MSE	PSNR
Proliferative DR	Sobel	12955.1471	7.0063
	Roberts	12955.3838	7.0063
	LoG	12949.3228	7.0083
	Prewitt	12955.1490	7.0063
	Canny	12939.2809	7.0117
	FOMSK	11443.9972	7.5450
	QRFODD	10769.2160	8.3172
	PLCRT	9102.3372	8.5393

exceptional generalization across diverse datasets and clinical imagery. The PLCRT retinal edge detection technology route is shown in Fig. 1.

According to the data in Table 1 and Fig. 2, the PLCRT operator demonstrates significant superiority in edge detection for Background DR retinal images. It achieves the lowest MSE of 8157.1510 among all the evaluated operators, indicating that its detected edges are the closest to the expert annotated ground truth. Additionally, the PLCRT attains the highest PSNR of 9.0154 dB, which is notably higher than those of traditional operators such as Sobel (7.0552 dB) and Canny (7.0608 dB), as well as advanced methods like FOMSK (7.8145 dB) and QRFODD (8.3172 dB). These results clearly demonstrate that the PLCRT provides superior image quality and exhibits higher accuracy in identifying fine vascular structures in retinal images.

Table 2 and Fig. 3 present the performance of various edge detection operators on retinal images exhibiting stellate maculopathy, while Table 3 and Fig. 4 report the results for Proliferative DR. For stellate

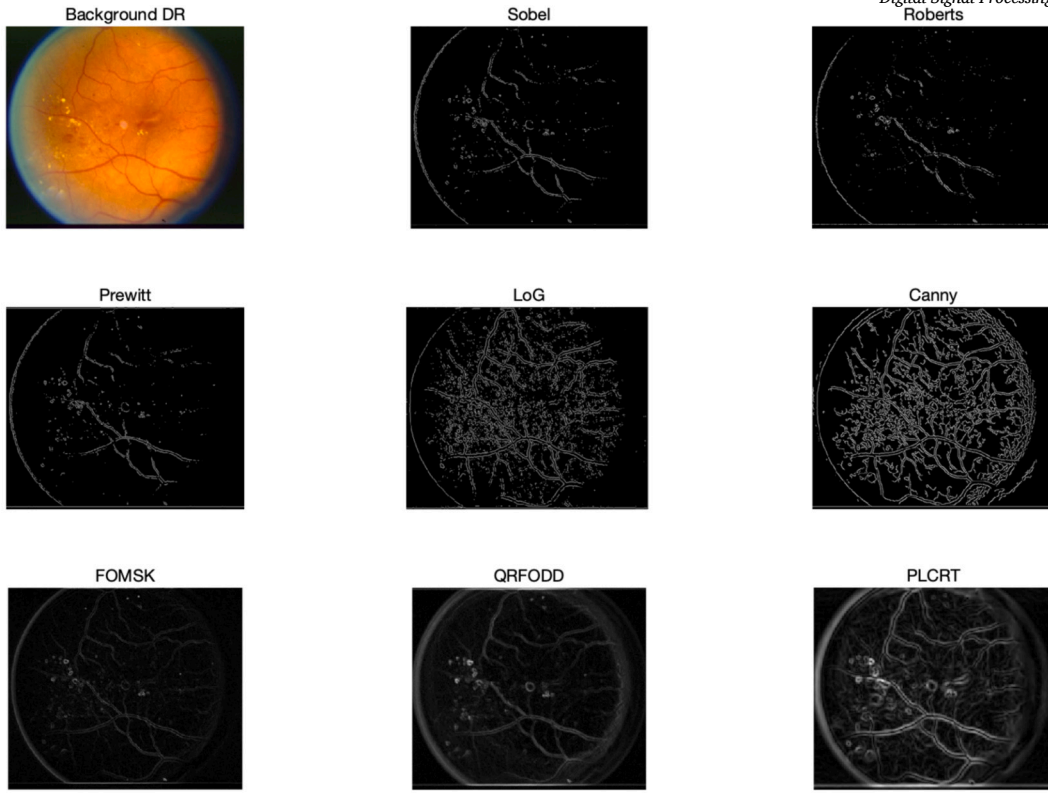


Fig. 2. Retinal lesion edge detection: the PLCRT for $M = (1, 2; 0, 1)$, $a = 1$, $\mathbf{b} = [1; 1]$, $\alpha = \frac{\pi}{6}$, $\gamma = \frac{\pi}{3}$, $\beta = \frac{5\pi}{6}$.

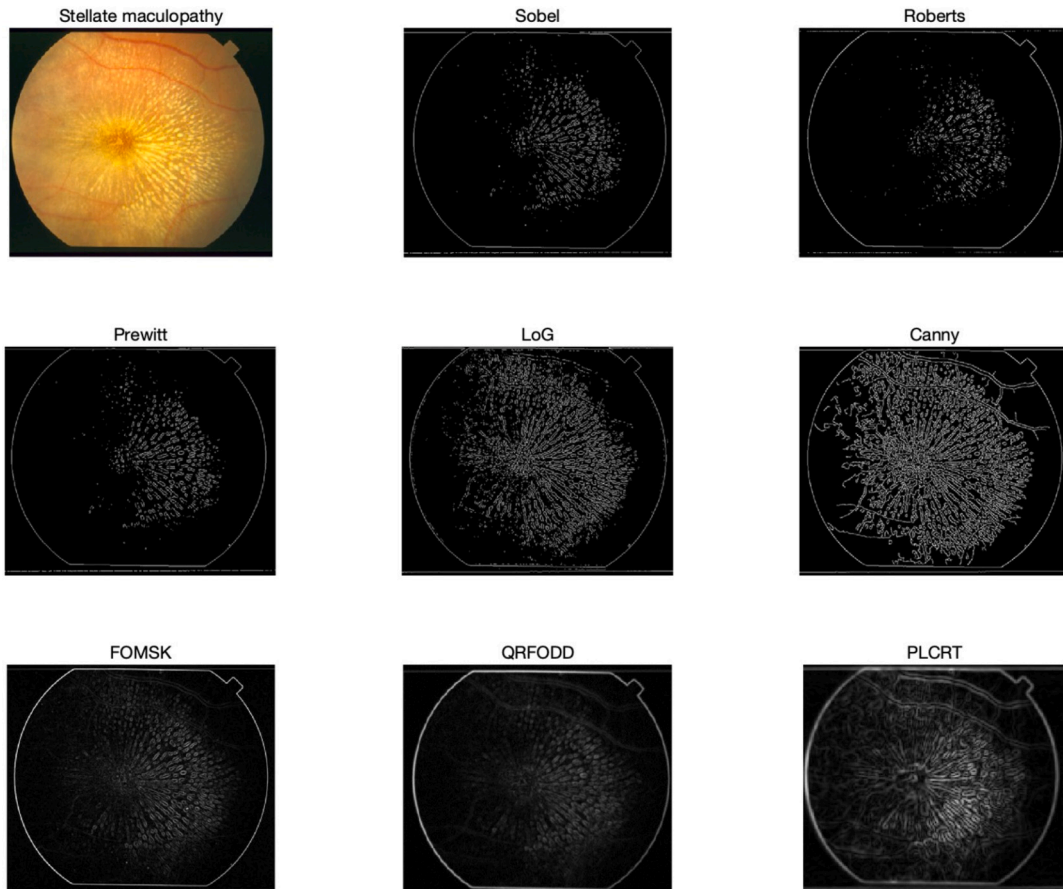


Fig. 3. Retinal lesion edge detection: the PLCRT for $M = (1, 2; 0, 1)$, $a = 1$, $\mathbf{b} = [1; 1]$, $\alpha = \frac{\pi}{6}$, $\gamma = \frac{\pi}{3}$, $\beta = \frac{5\pi}{6}$.

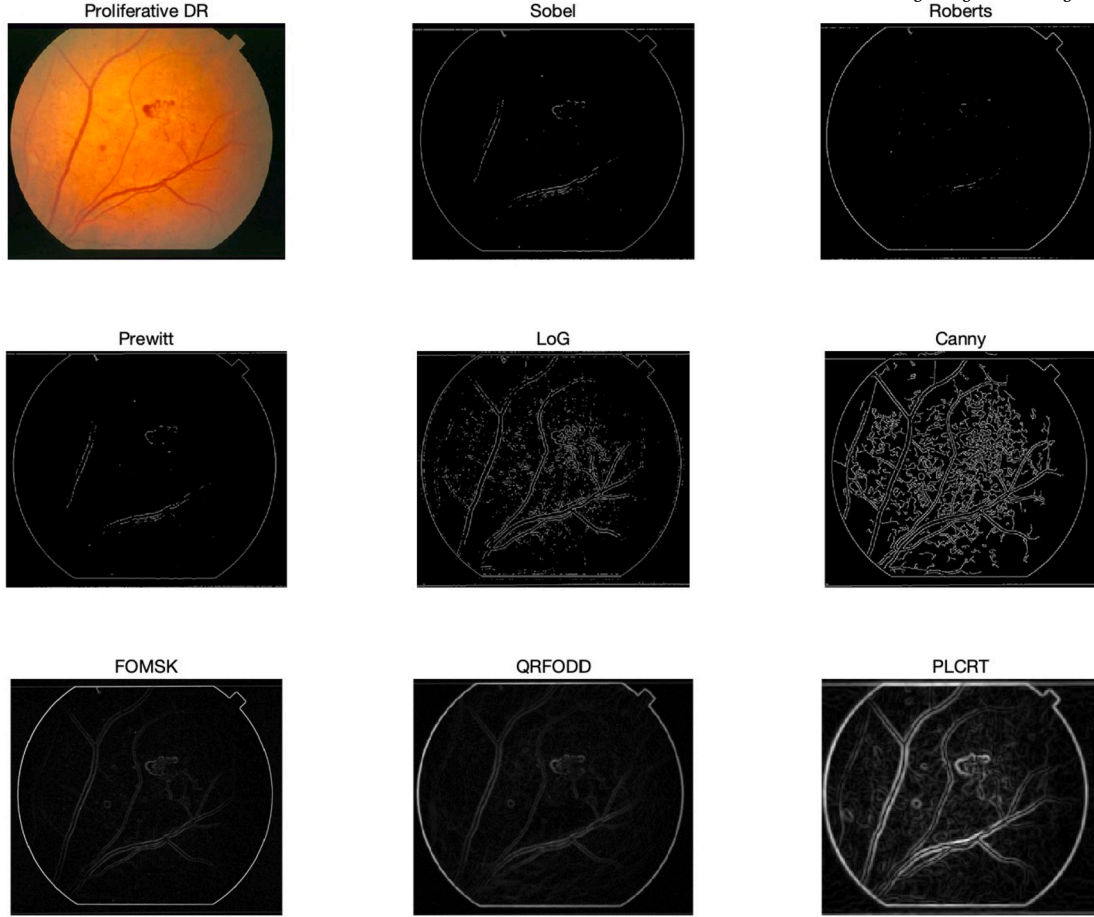


Fig. 4. Retinal lesion edge detection: the PLCRT for $M = (1, 2; 0, 1)$, $a = 1$, $\mathbf{b} = [1; 1]$, $\alpha = \frac{\pi}{6}$, $\gamma = \frac{\pi}{3}$, $\beta = \frac{5\pi}{6}$.

maculopathy, traditional operators such as Sobel, Prewitt, LoG, and Roberts yield comparable performance, with MSE values all exceeding 17,000 and PSNR values clustering around 5.82. These results indicate a limited capacity of conventional methods to extract edge information in complex lesion regions. Although the Roberts operator achieves a slightly lower MSE, its PSNR remains low, suggesting no meaningful improvement in image quality. Similarly, on Proliferative DR images, traditional operators produce MSE values around 12,900 and PSNR values consistently near 7.01, reflecting their shared limitation in accurately delineating neovascular and structurally complex pathological features. In contrast, the PLCRT operator demonstrates consistently superior performance across both pathological conditions. For stellate maculopathy, it reduces the MSE to 8745.3681 and raises the PSNR to 8.7130; for Proliferative DR, it lowers the MSE to 9102.3372 and increases the PSNR to 8.5393. These improvements outperform not only traditional methods but also recent advanced operators such as FOMSK (PSNR: 6.8557 and 7.5450, respectively) and QRFODD (PSNR: 6.9055 and 8.3172, respectively). Notably, the PLCRT significantly reduces edge detection error while enhancing fine structural details, demonstrating superior image enhancement capability.

It is well known that when the parameter matrix $M = (\cos\theta, \sin\theta; -\sin\theta, \cos\theta)$, the LCT degenerates into the fractional Fourier transform (FrFt) [27]. Fig. 5 illustrates the retinal edge detection results under this degenerate condition for two cases $\theta = \pi/6$ and $\theta = \pi/3$. The differences in detection performance are clearly visible, highlighting the significant impact of parameter selection on the effectiveness of FrFt-based edge detection.

Fig. 5 and Table 4 provide a comprehensive evaluation of the edge detection performance when the LCT degenerates into the FrFt, under

Table 4

Edge detection performance with the LCT reduced to the FrFt.

Retinopathy	$\theta = \pi/6$		$\theta = \pi/3$	
	MSE	PSNR	MSE	PSNR
Background DR	8721.4277	8.4452	8308.7289	8.8334
Stellate Maculopathy	10624.2838	7.8678	9120.7128	8.5305
Proliferative DR	9438.9969	8.3815	9934.9992	8.1591

two different fractional orders ($\theta = \pi/6$ and $\theta = \pi/3$). The results clearly demonstrate that FrFt achieves good edge clarity, structural continuity, and background suppression, with detection performance highly sensitive to the choice of θ . Specifically, $\theta = \pi/3$ yields better performance for Background DR and Stellate Maculopathy, enhancing lesion boundaries and texture details, while $\theta = \pi/6$ performs better for Proliferative DR, likely due to its ability to better capture fine, complex structures such as neovascularization. Compared to the full PLCRT method prior to degeneration, the FrFt offers a more lightweight formulation with reduced computational complexity and fewer parameters. However, it falls short in detection accuracy and structural fidelity. For example, in Background DR, the MSE of PLCRT is 8157.15, significantly lower than the best FrFt result of 8308.73 ($\theta = \pi/3$). Similarly, in Stellate Maculopathy, the PLCRT achieves a lower MSE (8745.37) than the FrFt (9120.71 at best). These findings indicate that although the FrFt can perform well with optimized parameters, its representational power and adaptability are inherently limited as a special case of the LCT. Therefore, the FrFt should be considered a practical intermediary technique rather than a full substitute for the PLCRT.

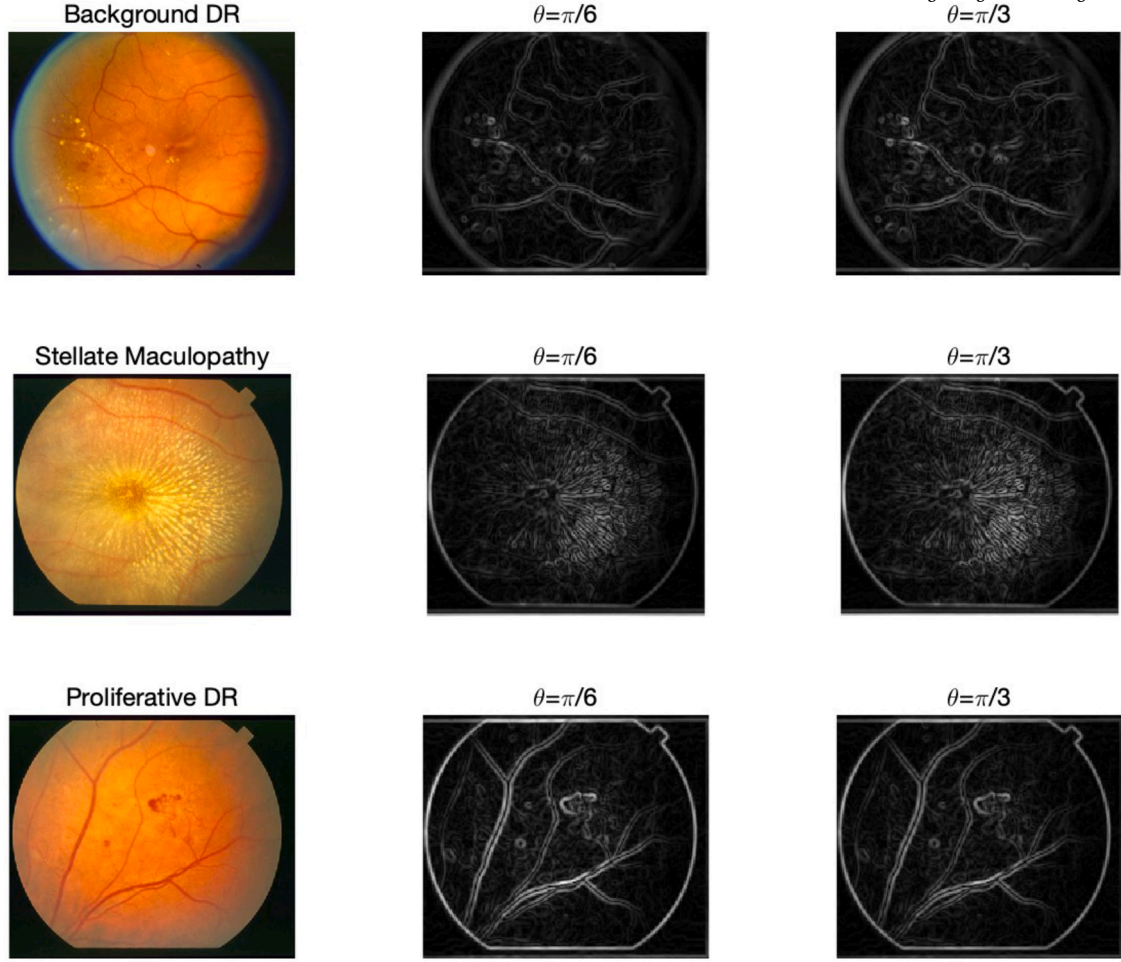


Fig. 5. Retinal lesion edge detection: the PLCRT for $M = (\cos\theta, \sin\theta; -\sin\theta, \cos\theta)$, $a = 1$, $\mathbf{b} = [1; 1]$, $\alpha = \frac{\pi}{6}$, $\gamma = \frac{\pi}{3}$, $\beta = \frac{5\pi}{6}$.

Table 5

Performance metrics of edge detection with added noise.

Retinopathy	Salt and pepper noise		Gaussian noise	
	MSE	PSNR	MSE	PSNR
Background DR	8641.7993	8.8562	10882.8576	7.7634
Stellate Maculopathy	10327.4583	7.9909	12614.7290	7.1220
Proliferative DR	10373.2792	7.9716	11670.4078	7.4599

Table 5 provides a quantitative evaluation of the PLCRT's edge detection performance under different noise conditions. Overall, the MSE values are lower and PSNR values higher under salt and pepper noise, indicating stronger suppression of impulsive noise by the PLCRT. Notably, Background DR images achieve the best results under both noise types, suggesting that their structural features are more easily captured. In contrast, the performance on Stellate Maculopathy and Proliferative DR images slightly declines under Gaussian noise, reflecting the greater challenge posed by their structural complexity.

Fig. 6 illustrates the edge detection performance of the PLCRT operator on retinal images contaminated with salt and pepper noise and Gaussian noise, respectively. The visual results indicate that, under impulsive noise interference, the PLCRT remains capable of accurately extracting vascular edges and lesion contours. The resulting edge maps demonstrate strong continuity and structural integrity, with minimal presence of false edges. Although Gaussian noise introduces noticeable blurring, the PLCRT still preserves clear edge structures and effectively delineates lesion regions from the background, highlighting its strong capability for detail preservation and adaptability. These visual outcomes clearly

Table 6

Comparison of complexity and computation time (image size $n = N \times M$).

Operator	Complexity	Computation time (s)		
		Background DR	Stellate maculopathy	Proliferative DR
Sobel	$\mathcal{O}(n)$	0.0487	0.0540	0.0329
Roberts	$\mathcal{O}(n)$	0.0436	0.0449	0.0344
LoG	$\mathcal{O}(n)$	0.0466	0.0391	0.0357
Prewitt	$\mathcal{O}(n)$	0.0021	0.0015	0.0016
Canny	$\mathcal{O}(n)$	0.1498	0.1314	0.1044
FOMSK	$\mathcal{O}(n)$	1.8808	1.9103	1.8952
QRFODD	$\mathcal{O}(n(\log n)^2)$	0.5107	0.6982	0.5649
PLCRT	$\mathcal{O}(n \log n)$	0.3127	0.3879	0.3627

demonstrate the robustness of the PLCRT in handling various types of noise while maintaining effective edge representation.

Table 6 provides a comparative analysis of complexity and actual computation time of various edge detection operators. The results demonstrate that the PLCRT operator maintains strong edge detection performance while offering significantly higher computational efficiency than advanced methods such as FOMSK and QRFODD. With a runtime of approximately 0.3–0.4 seconds, the PLCRT is markedly faster than the over 1.8 seconds required by the latter two. Although its computational complexity is higher than that of traditional operators, the PLCRT achieves a well balanced trade off between performance and efficiency, highlighting its practical utility particularly in medical image analysis tasks where both accuracy and processing speed are critical.

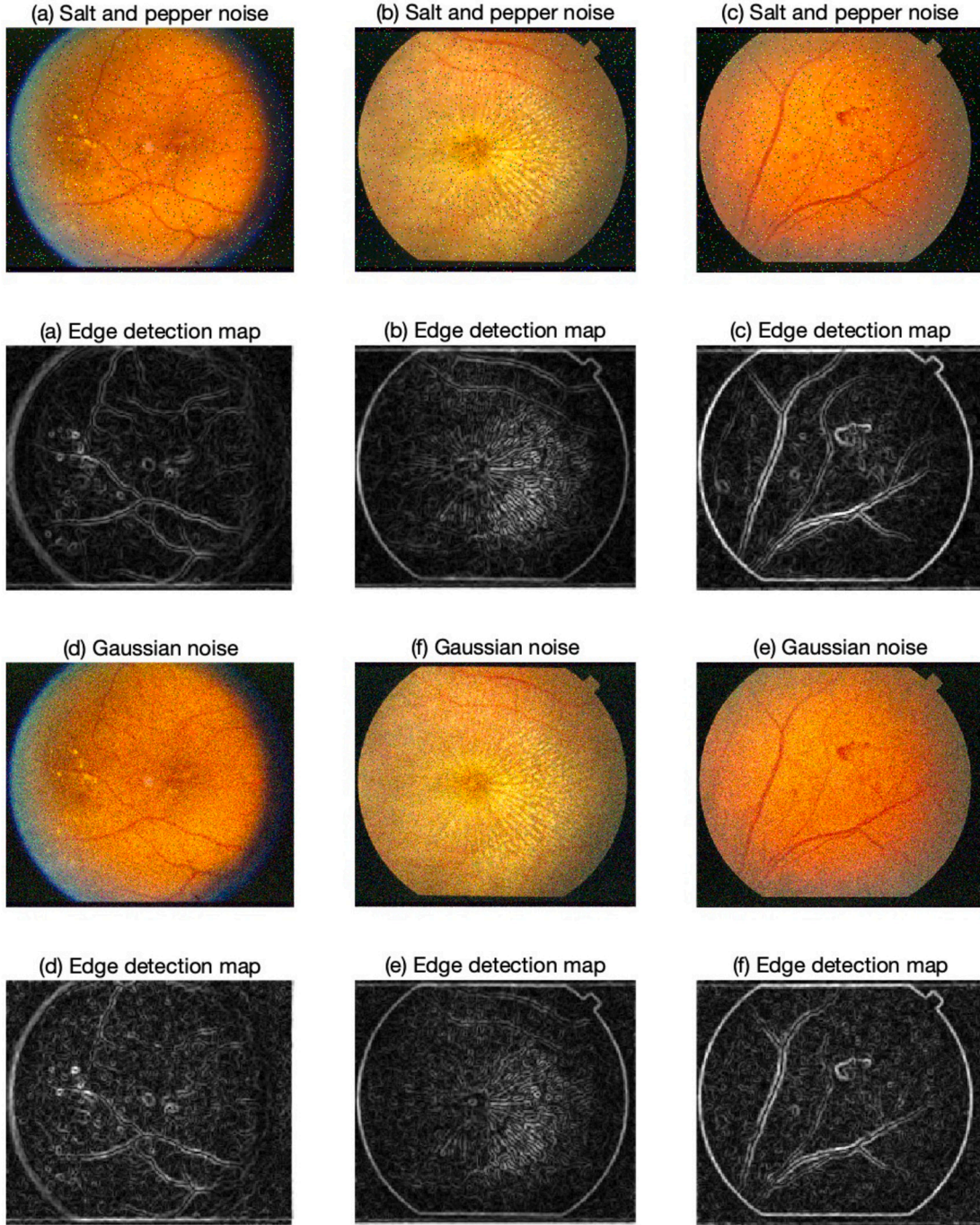


Fig. 6. Retinal lesion edge detection: the PLCRT for $M = (1, 2; 0, 1)$, $a = 1$, $\mathbf{b} = [1; 1]$, $\alpha = \frac{\pi}{6}$, $\gamma = \frac{\pi}{3}$, $\beta = \frac{5\pi}{6}$.

Fig. 7 presents the edge detection results of the PLCRT operator applied to clinically acquired retinal images from a hospital setting, aiming to evaluate its adaptability and applicability in real world clinical scenarios. As shown in the figure, the PLCRT is capable of clearly extracting the vascular network within the retina. This result indicates that the PLCRT not only performs well on standard datasets but also demonstrates stable and accurate edge detection capabilities in actual clinical images, highlighting its strong potential for practical medical applications.

5. Conclusion

The polar linear canonical ridgelet transform (PLCRT) is an innovative mathematical tool that integrates the Radon transform, ridgelet

transform, and the linear canonical transform (LCT), offering high degrees of freedom, multi-scale adaptability, and directional sensitivity. The PLCRT not only overcomes the limitations of traditional Fourier and wavelet transforms in capturing the directional features of high dimensional signals but also provides enhanced time-frequency localization capabilities, making it highly applicable in signal analysis, image processing, quantum physics, and optical transformations.

From a mathematical perspective, the PLCRT achieves greater flexibility in scale transformation, rotation, and projection by adjusting its transformation kernel and directional projections. From a geometric perspective, it efficiently represents the directional structure of high dimensional signals, improving its ability to extract curvilinear features and edge information. From a physical perspective, it extends the applicabil-

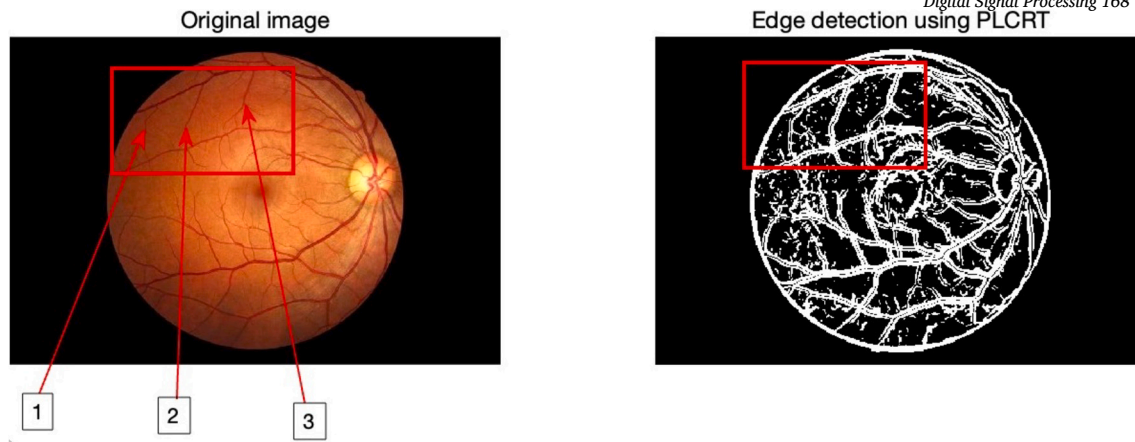


Fig. 7. Retinal lesion edge detection: the PLCRT for $M = (1, 2; 0, 1)$, $a = 1$, $\mathbf{b} = [1; 1]$, $\alpha = \frac{\pi}{6}$, $\gamma = \frac{\pi}{3}$, $\beta = \frac{5\pi}{6}$.

ity of Heisenberg's uncertainty principle, demonstrating its potential in quantum measurement, optical diffraction, and image reconstruction.

In summary, the PLCRT exhibits promising advantages in applications such as medical imaging (CT), speech enhancement, and remote sensing image analysis, potentially serving as a powerful tool for future signal processing technologies. Future research will further explore its potential applications in deep learning, pattern recognition, parameter optimization, and adaptive filtering.

CRediT authorship contribution statement

Rong-Qian Linghu: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft. **Bing-Zhao Li:** Conceptualization, Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in this paper.

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